IS5109 – IS PROJECT FOR COMMUNITY

PROJECT PROPOSAL



Medixmart Plus

GROUP 13

Department of Computing & Information Systems
Faculty of Computing
Sabaragamuwa University of Sri Lanka

September 2026

Approval of Community Project

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1. Title of the Project : Medixmart Plus
2. Name of the Internal Supervisor : Ms. T. Kayalvizhy
3. Internal Supervisor's Designation : Lecturer (Temp)

For office use only:

Approved/Not approved : Approved

Signature of the Internal Supervisor : T. Kayalvizhy

Date : 27/09/2026

Suggestions, if any :

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1. Acknowledgment

We want to express our deepest gratitude and appreciation to Prof. B.T.G.S Kumara, the Dean of the Faculty of Computing, Sabaragamuwa University of Sri Lanka, Mr. G.A.C.A. Herath, the Head of the Department of Computing and Information Systems, Faculty of Computing, Sabaragamuwa University of Sri Lanka and MS. A.W.T. Dulmini Lecturer (Temp), Community Project Coordinator, Department of Computing and Information systems, Faculty of Computing, Sabaragamuwa University of Sri Lanka for their coordination and high supervision of our Community project. Their guidance, support, and commitment have been instrumental in ensuring the successful execution of our project. We are grateful for their invaluable advice, encouragement, and expertise throughout the process. We also thank Ms. T. Kayalvizhy, Lecturer (Temp) and Internal Supervisor of the Community Project.

We appreciate their valuable insights, technical expertise, and continuous support. We also thank all faculty members, staff, and fellow students who provided valuable feedback, resources, and assistance throughout this project. We greatly appreciate their input and involvement. Lastly, we thank our friends and family for their unwavering support, understanding, and encouragement throughout this endeavor. Their belief in our abilities has been a constant source of motivation. We are truly grateful for their help and guidance.

2. Introduction and Objectives

2.1 Introduction

The selected community partner is a pharmacy located near the university. During the initial interview, the pharmacy staff explained that several important activities are handled manually. The pharmacy maintains medicine stock records manually, making it difficult to quickly identify current medicine quantities. The pharmacy also needs to monitor expiry dates because some medicines can be returned to suppliers approximately seven months before their expiry, according to the pharmacy's current practice.

Another identified issue is the absence of an automatic method for identifying medicines that are reaching a low stock level. In addition, the pharmacy needs a practical way to identify customers who obtain medicines on credit and keep track of their outstanding amounts. The pharmacy also suggested that automated customer messages would be useful for reminders and other relevant communications.

To address these problems, the proposed Medixmart Plus system will provide a mobile and Desktop application digital solution for medicine and stock management, expiry and return alerts, low-stock alerts, customer credit management, and customer notifications. The system will provide a dashboard and reports so that pharmacy staff can access important information quickly and reduce dependence on manual records.

The project will be developed based on the actual requirements gathered from the community partner. The proposed system is intended to support pharmacy staff rather than replace professional pharmacy decision-making. Medicine dispensing and clinical decisions will remain the responsibility of qualified pharmacy personnel.

2.2 Objectives

Overall Objective

To develop a user-friendly mobile and Desktop application pharmacy management system that reduces manual work and improves the management of medicine stock, expiry/return dates, and customer credit information.

Specific Objectives

- To automate the current manual Stock Management process for medicines.
- To automatically update medicine stock quantity based on medicine sales.
- To keep different stock records when the same medicine has a different expiry date.
- To give an alert for medicines expiring seven months in advance.
- To identify medicines when their available stock becomes low.
- To notify the staff of the pharmacy if they have low stock.
- To keep suppliers up to date on information.
- To enter details about medicines received and ordered.
- To keep basic information about the customers.
- To detect customers' birthdays and offer birthday wishes.
- To supply valuable reports on medicine stock, sales, expiry dates, and low-stock medicines.

3. Analysis

3.1 Feasibility Study

The feasibility study evaluates whether the proposed Medixmart Plus system can be successfully developed and implemented within the available technical, operational, and economic limitations.

3.1.1 Technical Feasibility

The proposed system is technically feasible because it can be developed using commonly available technologies, mobile devices, and desktop computers. A smartphone, tablet, or desktop computer can be used by the pharmacy staff to access the system. The application can communicate with a centralized cloud database, processing all business logic separately on the client side in the mobile and desktop applications.

The proposed technologies include:

- Flutter for mobile application development
- Node.js and Express.js for backend development
- PostgreSQL for database management
- REST API for communication between the mobile application and server
- Firebase Cloud Messaging or a similar service for application notifications
- GitHub for source-code management
- Figma for interface design

These technologies are widely available and provide sufficient functionality to implement the proposed system. The development team also has access to computers, smartphones, internet facilities, and free development environments required to develop and test the system.

3.1.2 Operational Feasibility

The system is operationally feasible because it is designed according to requirements obtained directly from MedixMart Pharmacy. Currently, employees perform their work manually. The proposed system will simplify these activities rather than introduce unnecessary additional processes.

The mobile application will provide a simple and user-friendly interface that allows staff to:

- Create bills
- Search medicines
- Add stock
- Update stock
- Check expiry dates
- View alerts

- Maintain supplier records
- Maintain customer information
- Generate reports

Since the pharmacy prefers using mobile devices, a mobile-based solution should be convenient for its daily operations. Basic training and user guidance will be provided to the pharmacy staff when the completed system is handed over.

3.1.3 Economic Feasibility

The proposed project is economically feasible because most of the software and development tools required are free or open source. Flutter, Node.js, PostgreSQL, Visual Studio Code, GitHub, and Figma provide free options suitable for developing the proposed system. The pharmacy can initially use an existing smartphone, reducing the requirement for additional hardware.

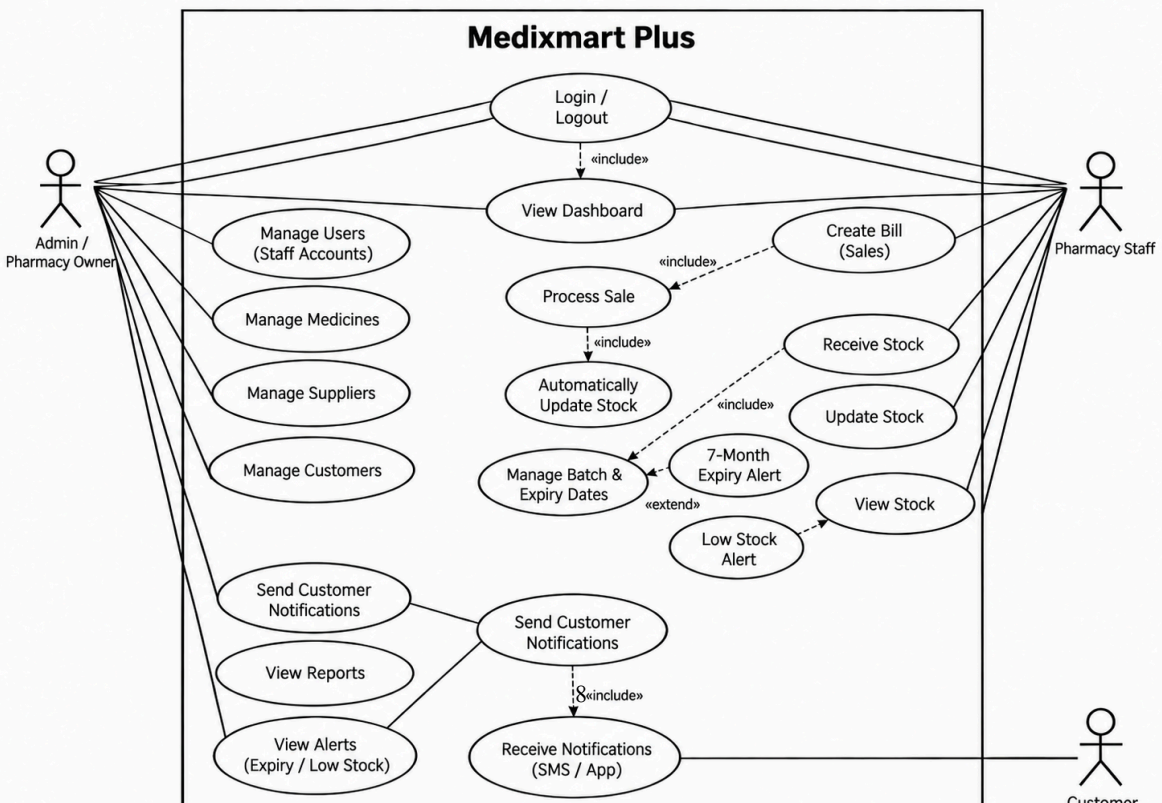
Possible costs may include:

- Internet/data usage
- Cloud hosting if required
- Optional SMS services
- Printing and documentation
- Transportation for organization visits

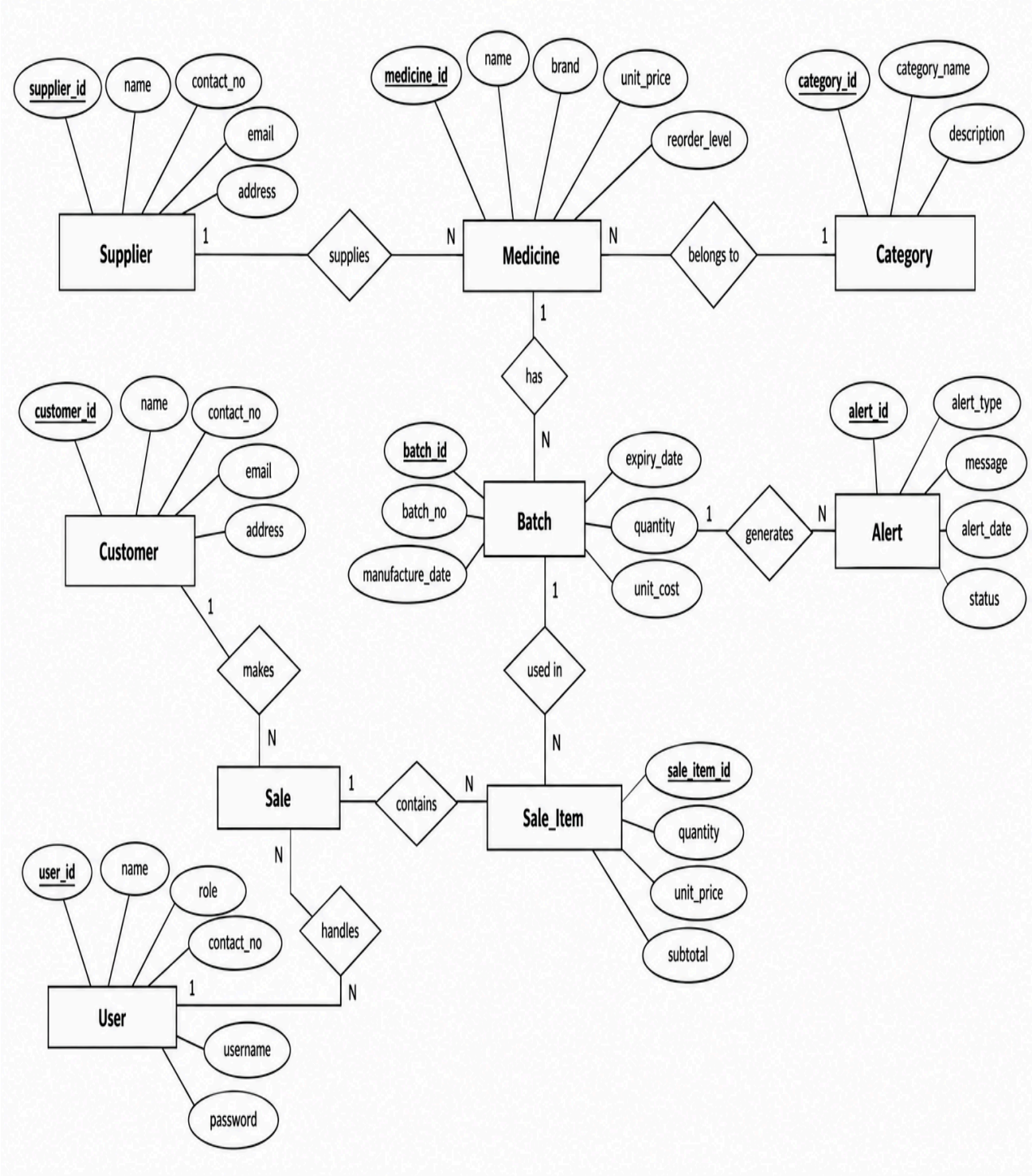
Therefore, the initial development cost of the system is expected to be relatively low.

3.2 Diagrams

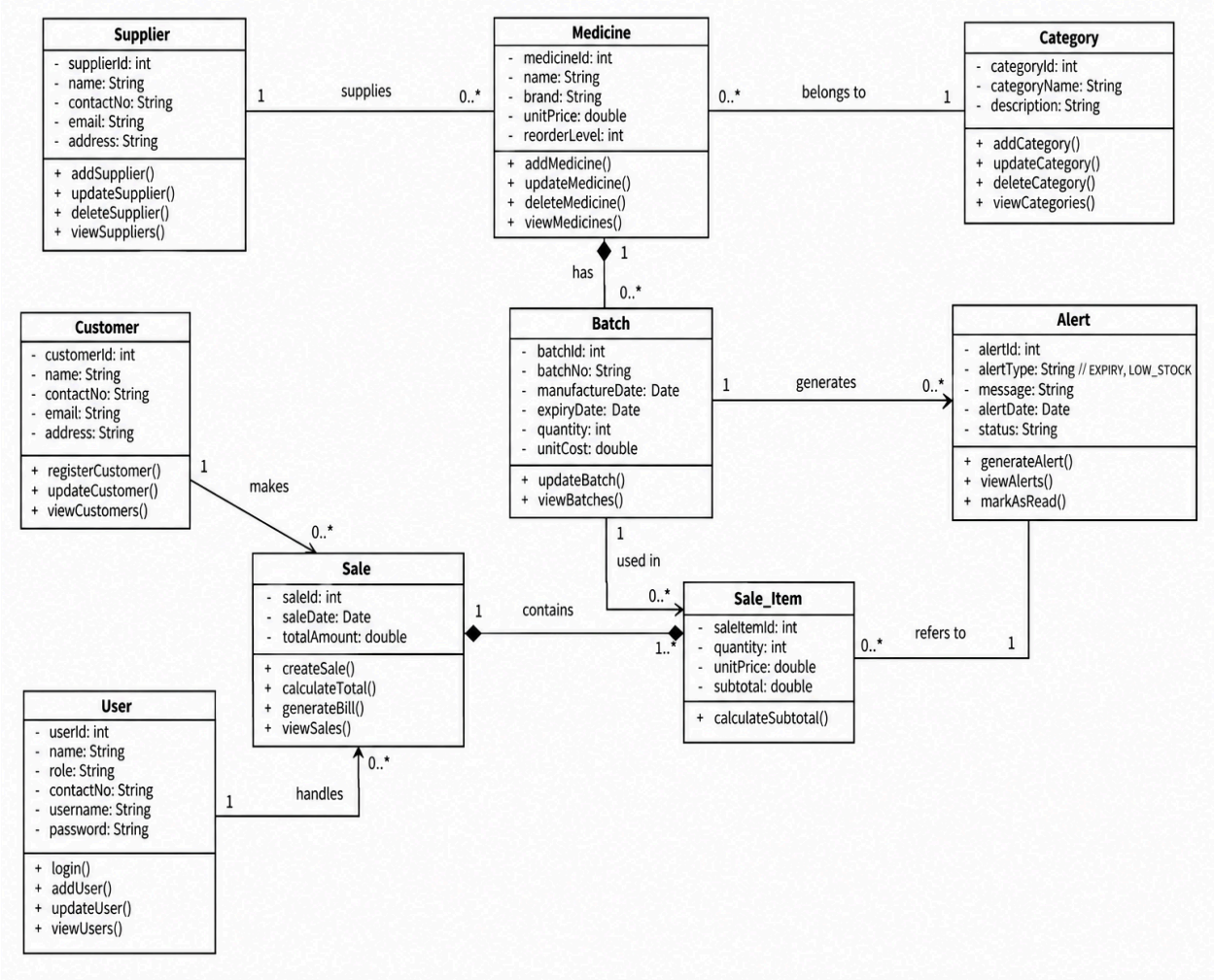
3.2.1 Use Case Diagram



3.2.2 ER Diagram



3.2.3 Class Diagram



4. Development Environment

4.1 Hardware Requirements

Hardware	Minimum Requirement
Developer laptop/PC	Intel Core i3/Ryzen 3 or above
RAM	4 GB minimum; 8 GB recommended
Storage	10 GB free space for development
Display	1366 × 768 or above
Internet	Required for package installation, Git, and optional messaging APIs
Smartphone	Optional for responsive testing and notification testing

4.2 Software Requirements

- Windows/Linux/macOS operating system
- Visual Studio Code
- Git and GitHub
- Node.js and npm
- Python 3.x and pip
- PostgreSQL
- Modern web browsers such as Google Chrome or Microsoft Edge
- Postman
- draw.io for UML/ER diagrams

4.3 Technology Requirements

Layer	Technology
Frontend	Flutter (mobile/desktop), Electron / C# .NET (desktop)
Backend	Node.js with Express.js
Database	PostgreSQL
Version Control	Git + GitHub
API	REST API
Notification Technology	Firebase Cloud Messaging may be used for application notifications.
Deployment	Local server / suitable cloud hosting
Testing	Postman

5. Tables and Structure

5.1 Number of Modules

No.	Module
01	User & Authentication Management
02	Medicine & Category Management
03	Stock & Purchase Management
04	Expiry & Return Alert Management
05	Sales & Credit Management
06	Customer Management & Notifications
07	Dashboard & Reports
08	System Administration

5.2 Module Details

Module	Main Functions
User & Authentication	Login, logout, password change, role-based access
Medicine & Category	Add/update/search medicines, categories, prices, and reorder levels.
Stock & Purchase	Record purchases, batches, quantities, and suppliers; update stock
Expiry & Return	Track batch expiry dates and generate return-soon/expired alerts
Sales & Credit	Record sales, credit transactions, payments, and outstanding balances
Customer & Notifications	Customer profiles, credit customer identification, and approved reminders
Dashboard & Reports	Low-stock count, return-soon count, expired count, credit balances, and reports.
Administration	Manage users, settings, backup/export, and audit-related information.

6. Proposed System

6.1 Requirements

6.1.1 Functional Requirements

FR01 - User Login
The system shall allow authorized pharmacy staff to log in using valid user credentials.

FR02 - Medicine Management
The system shall allow authorized users to add, edit, search, and view medicine information.

FR03 - New Stock Recording
The system shall allow pharmacy staff to record newly received medicine stock.

FR04 - Stock Details
The system shall maintain medicine quantity, received date, expiry date, and associated supplier information.

FR05 - Different Expiry Dates
The system shall maintain separate stock records when the same medicine has different expiry dates.

FR06 - Sales Total Calculation
The system shall calculate the total amount of a sales transaction.

FR07 - Automatic Stock Update
The system shall automatically reduce the available medicine quantity after a sales transaction is completed.

FR08 - Expiry Monitoring
The system shall monitor medicine expiry dates.

FR09 - Seven-Month Expiry Alert
The system shall generate an alert seven months before a medicine expires.

FR10 - Expired Medicine Identification
The system shall identify medicines that have passed their expiry dates.

FR11 - Low-Stock Monitoring
The system shall monitor available medicine quantities.

FR12 - Low-Stock Alert
The system shall notify pharmacy staff when a medicine reaches or falls below its defined minimum stock level.

FR13 - Supplier Management
The system shall allow users to add, update, search, and view supplier information.

FR14 - Customer Management
The system shall allow pharmacy staff to add, update, search, and view basic customer information.

FR15 - Customer Date of Birth
The system shall allow the date of birth of registered customers to be recorded.

FR16 - Birthday Reminder
The system shall identify registered customers whose birthday occurs on the current date.

FR17 - Birthday Wish Support
The system shall support pharmacy staff in providing birthday wishes to customers.

FR18 - Dashboard
The system shall display important stock, expiry, sales, and customer reminder information through a dashboard.

FR19 - Reports
The system shall generate stock, sales, expiry, and low-stock reports.

FR20 - Logout
The system shall allow users to securely log out.

6.1.2 Non-Functional Requirements

1. Usability

The system should be easy to use and user-friendly for the pharmacy personnel.

2. Performance

The system should allow for satisfactory response times in conducting normal activities like searching for medicines, recording sales, checking stock, etc.

3. Reliability

It should correctly record medicine, stock, sales, supplier, and customer information.

4. Availability

The system shall be available upon request during the Pharmacy operating hours.

5. Maintainability

The system should be modular in nature so that future changes can be made.

6. Scalability

The system should be able to accommodate more medicines, stock, sales, suppliers, and customer records.

7. Compatibility

It should work on supported desktop operating systems and mobile devices.

8. Data Accuracy

The system should check for meaningful information before it is stored.

9. Backup

The system should back up the database frequently to minimize the chance of permanent data loss.

6.1.3 Security Requirements

- Only authorized users should be allowed to access the system.
- Users will be asked to log on with a username and password.
- Passwords will not be kept in clear text.
- Passwords will be stored in a secure hash.
- Any protected pages should not be accessible without authentication.
- Where appropriate, access to users can be limited by assigned roles.
- Before processing or storing user inputs, they should be validated.
- Only authorized personnel in the pharmacy will be able to access customer information.
- Access to the database should be restricted to prevent unauthorized access.
- Frequent backups should be made.
- Activities of the system may be logged for accountability and troubleshooting purposes.

6.2 Uniqueness of the Product

Medixmart Plus is designed according to the actual operational requirements identified at MedixMart Pharmacy rather than being a general-purpose inventory application.

The proposed system combines several important pharmacy activities within a single mobile-based platform:

- Digital billing
- Automatic stock updating
- Batch-based medicine management
- Seven-month expiry alerts
- Low-stock alerts
- Supplier management
- Customer management
- Customer greeting and notification facilities
- Pharmacy reports

A major feature of the system is its batch-based expiry management. Different batches of the same medicine can be stored separately with their respective received dates, quantities, suppliers, and expiry dates.

The seven-month advance expiry warning is also specifically based on the requirements provided by MedixMart Pharmacy.

6.3 Methodology (Agile Methodology)

The system will be created with the help of Agile Software Development Methodology.

The reason why Agile is appropriate for this community project is that the system can be built incrementally and evaluated regularly with the pharmacy stakeholder.

Phase 1 - Requirement Gathering

The team will get to know how the manual process works at Medixmart Pharmacy and come up with the primary issues and needs.

Phase 2 - Requirement Analysis

The data collected will be analyzed to establish the primary functions of the system and scope of the project.

Phase 3 - System Design

The team will prepare,

- Use Case Diagram
- ER Diagram
- Class Diagram
- Database design
- User-interface designs

Phase 4 - Development

The system will be designed in a modular fashion.

The suggested development sequence is that they can access authentication, medicine management, stock management, alerts, suppliers, customers, and dashboards and reports.

Phase 5 - Testing

All modules will be tested separately. Once the modules are complete, they will undergo integration testing to ensure that all of them function properly in conjunction with one another.

Phase 6 - Stakeholder Feedback

A system prototype will be demonstrated to Medixmart Pharmacy.

Feedback will be collected on,

- Usability
- Functions
- Required changes
- Accuracy
- Practical use

Phase 7 - Improvements

Improvements required will be made as per stakeholder/ supervisor's feedback.

Phase 8 -Final Testing and Delivery

The final tests will be conducted on the completed system before demonstration and delivery.

7. Gantt Chart

Activity	Time Duration (Weekly)														
	1st Month				2nd Month				3rd Month				4th Month		
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Problem & Organization Selection															
Requirement Gathering															
Feasibility & Requirement Analysis															
Proposal / SRS Preparation															
System Design															
System Development															
Testing & Stakeholder Feedback															
System Improvements															
Final Product & Documentation															
Final Report, Presentation & Handover															

8. Cost Estimate

The proposed Medixmart Plus system will mainly be developed using free and open-source technologies. Therefore, the software development cost is expected to be relatively low. The main expenses of the project will be related to internet usage, transportation, documentation, and deployment.

Item	Estimated Cost (LKR)
Desktop Application Framework	Free
Node.js and Express.js	Free
PostgreSQL	Free
Visual Studio Code	Free
Git and GitHub	Free
Figma	Free
Draw.io	Free
Existing Development Computers	No additional cost
Existing Computer at Pharmacy	No additional cost
Transportation for Pharmacy Visits	Rs. 1,000
Testing and Miscellaneous Expenses	Rs. 1,500
Optional Hosting / Deployment	Rs. 0
Total Estimated Cost	**Rs. 2,500**

9. Conclusion

MedixMart Pharmacy currently performs important operations such as billing, stock recording, stock updating, supplier record management, and expiry monitoring manually. This creates difficulties in maintaining accurate and easily accessible information and increases the effort required for routine pharmacy operations.

The proposed Medixmart Plus - Mobile and Desktop Application Pharmacy Inventory, Billing, Expiry Alert and Customer Management System aims to digitalize these activities within one integrated platform.

The system will support digital billing, automatic stock updating, batch-based inventory management, seven-month expiry alerts, low-stock notifications, supplier management, customer management, customer notifications, and reporting facilities.

Since the proposed system is based on requirements directly identified through discussions with MedixMart Pharmacy and is designed for mobile use according to their preference, the project is expected to provide a practical solution for improving the efficiency and accuracy of the pharmacy's daily operations.

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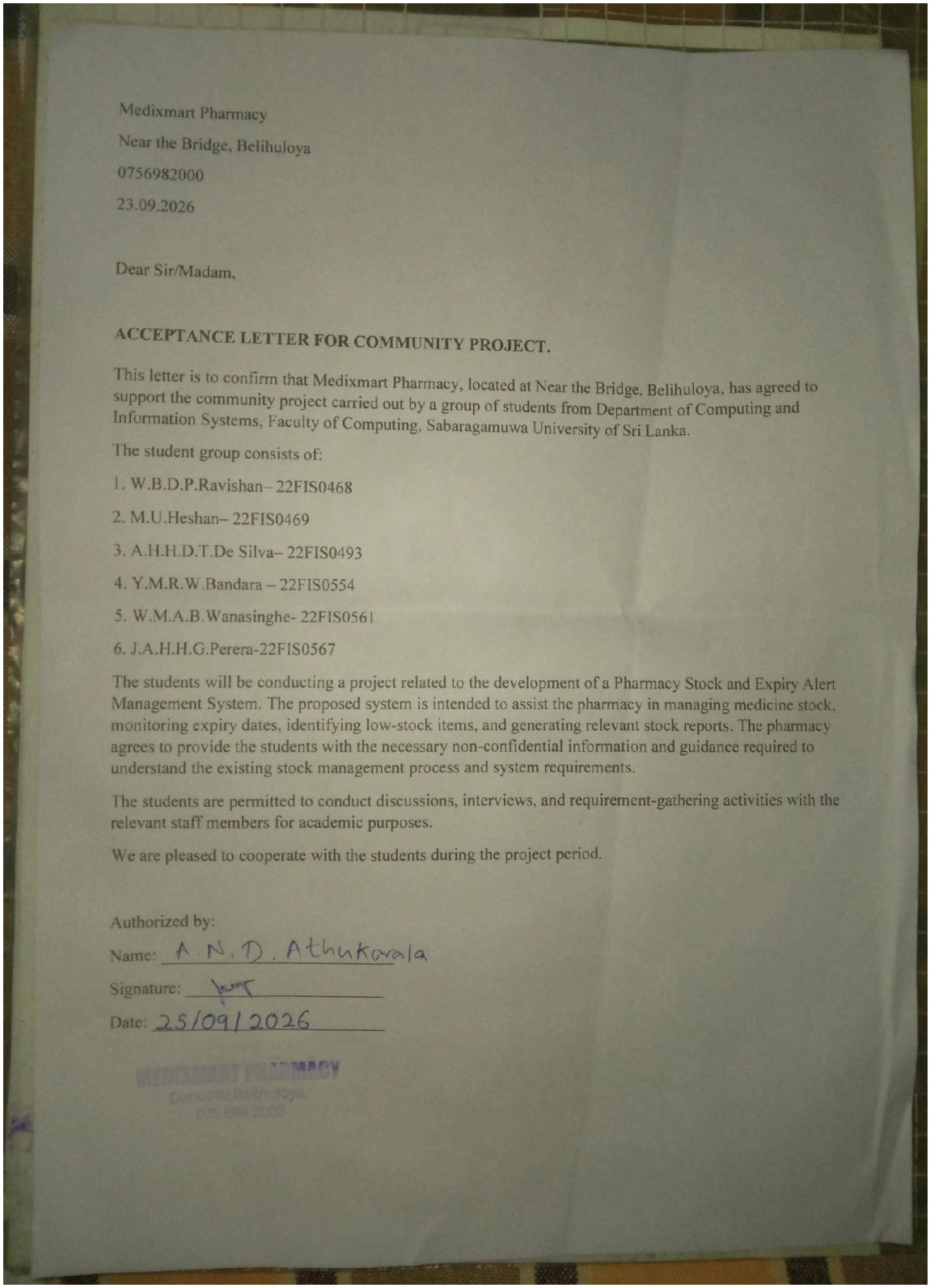
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11. Appendix

Include the approval letter obtained from the organization for conducting the project.



Medixmart Pharmacy

Near the Bridge, Belihuloya

0756982000

23.09.2026

Dear Sir/Madam,

ACCEPTANCE LETTER FOR COMMUNITY PROJECT.

This letter is to confirm that Medixmart Pharmacy, located at Near the Bridge, Belihuloya, has agreed to support the community project carried out by a group of students from Department of Computing and Information Systems, Faculty of Computing, Sabaragamuwa University of Sri Lanka.

The student group consists of:

1. W.B.D.P.Ravishan– 22FIS0468
2. M.U.Heshan– 22FIS0469
3. A.H.H.D.T.De Silva– 22FIS0493
4. Y.M.R.W.Bandara – 22FIS0554
5. W.M.A.B.Wanasinghe- 22FIS0561
6. J.A.H.H.G.Perera-22FIS0567

The students will be conducting a project related to the development of a Pharmacy Stock and Expiry Alert Management System. The proposed system is intended to assist the pharmacy in managing medicine stock, monitoring expiry dates, identifying low-stock items, and generating relevant stock reports. The pharmacy agrees to provide the students with the necessary non-confidential information and guidance required to understand the existing stock management process and system requirements.

The students are permitted to conduct discussions, interviews, and requirement-gathering activities with the relevant staff members for academic purposes.

We are pleased to cooperate with the students during the project period.

Authorized by:

Name: A.N.D. Athukorala

Signature: _____

Date: 25/09/2026

MEDIXMART PHARMACY
Dimuthu, Belihuloya.
075 698 2000